

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

19EE701 - COMPUTER TECHNIQUES IN POWER SYSTEM ANALYSIS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I**Marks BT* CO* PO***

1. a) For the network shown in Fig. Q.1(a), obtain the incidence matrices **A**, **B**, **C** and **K**. Hence show that $\mathbf{A}_l \mathbf{K}^t = \mathbf{B}_l$. Take elements 1,2,3,4 as branches and elements 5,6,7 as links.

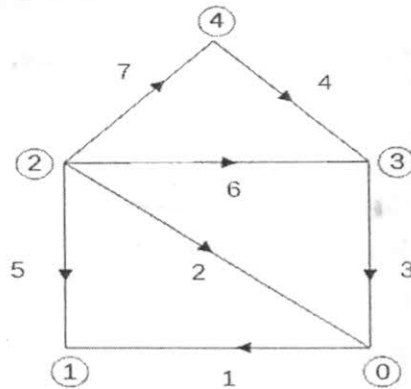


Figure Q1.(a)

10 L*2 1 1

- b) Define:
(i) graph
(ii) oriented graph
(iii) tree
(iv) co-tree.

4 L2 1 1

- c) Obtain bus incidence matrix **A** and node element incidence matrix $\hat{\mathbf{A}}$ from the matrix $\mathbf{A}^T$ given in table Q. 1(c). Hence draw the oriented graph for the same. Note: P,Q,R,S are buses and 1-7 are elements

$$\mathbf{A}^T = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} P \\ Q \\ R \\ S \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 1 \\ -1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix} \end{matrix}$$

Table Q 1(c)

6 L2 1 1

2. a) Define primitive network. Derive the expression for forming the bus admittance matrix from the primitive admittance matrix through singular transformation.

10 L2 2 1

- b) For the 3-bus network given in Figure Q 2(b) obtain bus admittance matrix by the method of singular transformation. The primitive admittance of different elements are marked on the figure in per unit. Note that the reference is ground, which is not shown in the figure.

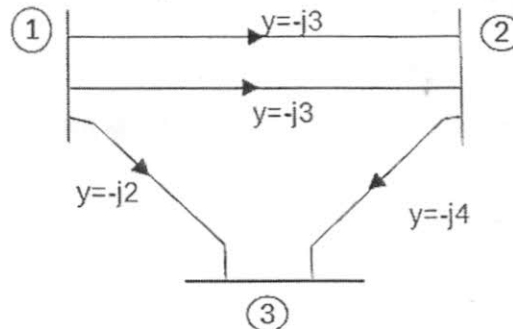


Figure Q 2(b)

- c) Derive the formula for addition of branch in Z bus building algorithm

6	L2	1,2	1
4	L1	2	1,2

3. a) Form the bus admittance matrix by method of inspection for the system shown in Fig. Q.3(a). The primitive impedances of the lines are marked on the figure in per unit. Note that the nodes 1,2,3,4 are all buses and the reference node is ground (which is not shown in the figure).

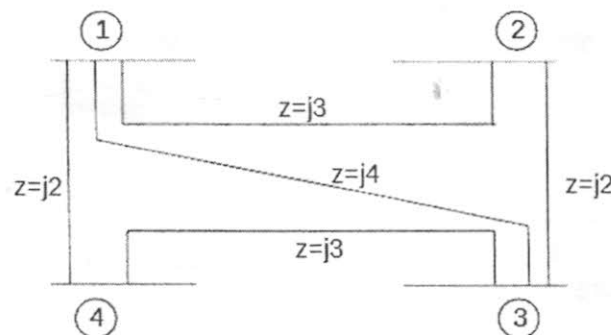


Fig. Q.3(a).

8	L2	2	1
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- b) For the network shown in Fig. Q.3(b), obtain Z_{bus} by Z_{bus} building algorithm taking node 1 as the reference. Self impedances in p.u. are marked on the diagram. Add elements in the ascending order of their numbering.

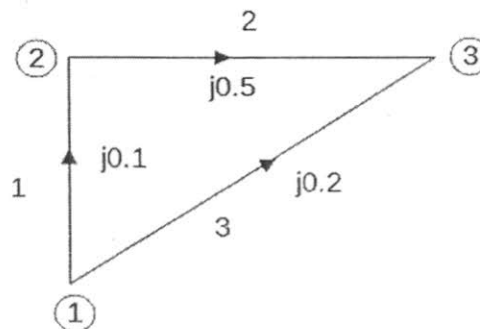


Fig. Q.3(b)

12	L3	2	1
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Unit – II

4. a) Discuss the advantages of using bus admittance matrix in load flow analysis.

4	L2	3	1
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- b) Describe the classification of buses for load flow analysis.
- c) Obtain the Gauss-Seidel load flow solution at the end of first iteration for the power system shown in Fig. Q. 4(c). Assume flat start for bus voltages V_3 and V_4 .

6 L1 3 1

SB	EB	R (p.u.)	X (p.u.)
1	2	0.05	0.15
1	3	0.1	0.3
2	3	0.15	0.45
2	4	0.1	0.3
3	4	0.05	0.15

Bus no.	Pi (p.u.)	Qi (p.u.)	Vi (p.u.)
1			1.04 angle 0
2	0.5	-0.2	
3	-1	0.5	
4	0.3	-0.1	

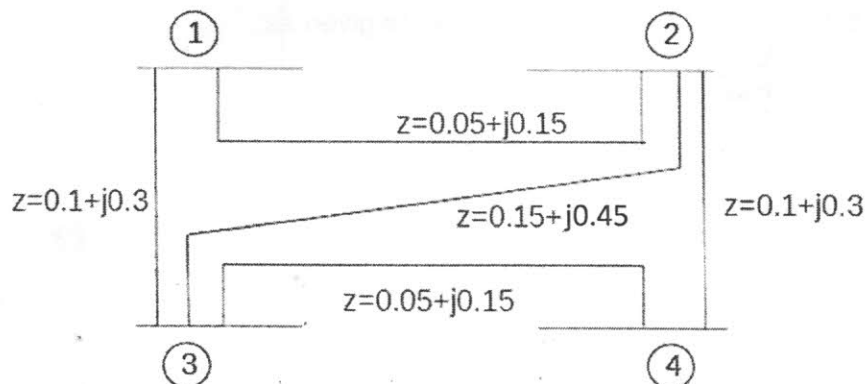


Fig. Q. 4(c).

10 L3 3 2

5. a) Obtain the voltages at all buses for the three-bus system shown in Fig. Q 5(a) at the end of first iteration by Newton-Raphson method. The data are given below.

Line data

SB	EB	R (p.u.)	X (p.u.)
1	2	0	0.1
1	3	0	0.2
2	3	0	0.2

Bus data

Bus no.	P_G (p.u.)	Q_G (p.u.)	P_L (p.u.)	Q_L (p.u.)	V (p.u.)
1					1 at angle 0.
2	5.3217				1.1
3			3.6392	0.5339	

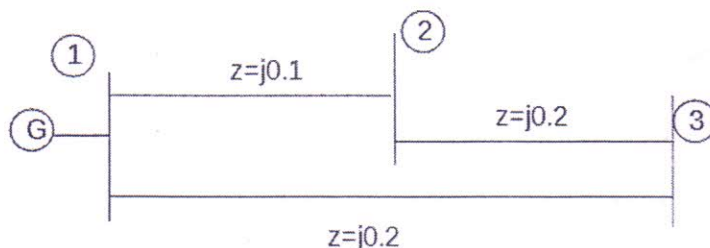


Fig. Q 5(a)

12 L4 3 2

- b) Compare Gauss-Seidel and Newton-Raphson methods of load flow analysis.

8 L2 3 1

6. a) Derive equal incremental cost criterion for economic generation scheduling, neglecting transmission losses. Also write step-by-step algorithm for obtaining economic loading of generators.

10 L2 4 2

- b) The fuel cost functions in \$/hour for three thermal units are given as:

$$F_1 = 350 + 7.2 P_{G1} + 0.004 P_{G1}^2$$

$$F_2 = 500 + 7.3 P_{G2} + 0.0025 P_{G2}^2$$

$$F_3 = 600 + 6.74 P_{G3} + 0.003 P_{G3}^2$$

P_{G1} , P_{G2} and P_{G3} are in MW. Neglecting transmission losses, find the optimal schedule. Compare the cost thus obtained to the case when generators share the load equally if demand P_D is 450 MW.

10 L3 4 2

Unit – III

7. a) Describe the solution of the swing equation by point-by-point method.

10 L2 5 1

- b) For the swing equation, $0.0331 \frac{d^2\delta}{dt^2} = 0.8 - 1.333 \sin\delta$, taking the starting instant as 0.15 second and step size as 0.05 second, calculate δ and ω at 0.20 second. Take the initial values as: $\delta_0 = 0.761$ rad, $\omega_0 = 2.072$ rad/s. Use modified Euler's method.

10 L4 5 2,5

8. a) Elaborate Runge-Kutta method of solving the swing equation.

10 L2 5 2

- b) With usual notations, explain representation of synchronous machines and loads for transient stability studies.

10 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
